

GridTokenX: A Decentralized Physical Infrastructure Network for Real-Time Energy Trading

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Abstract

This whitepaper introduces GridTokenX, a Decentralized Physical Infrastructure Network (DePIN) ([Messari, 2023](#)) purpose-built for real-time, peer-to-peer energy trading in distributed power grids. The global proliferation of rooftop solar installations, battery energy storage systems (BESS), and electric vehicle (EV) chargers has created millions of prosumers — participants who simultaneously produce and consume electricity — yet existing grid management infrastructure remains fundamentally centralized, opaque, and ill-suited to coordinate this distributed flexibility at scale.

GridTokenX addresses this structural gap by constructing a vertically integrated technology stack that spans from physical hardware to on-chain settlement. At the edge, industrial-grade IoT gateways translate raw meter telemetry from DLMS/COSEM and OCPP 2.0.1 protocols into cryptographically signed data payloads. These payloads traverse a mutual TLS (mTLS) secured service mesh, are validated by an Oracle Bridge microservice, and are ultimately committed to the Solana blockchain ([Yakovenko, 2018](#)) via Anchor-based smart contracts ([Coral, 2021](#)) — achieving end-to-end latency well below one second.

The platform's economic model is anchored by a triple-token architecture: **GRID**, a 1:1 kWh-pegged energy token minted upon verified renewable generation; **GRX**, a utility and governance token used for staking, fee payment, and protocol voting; and **gTHB**, a reserve-backed stablecoin pegged to the Thai Baht for stable, fiat-denominated settlement. A Continuous Double Auction (CDA) matching engine, implemented both off-chain for low-latency order matching and on-chain for trustless settlement, enables prosumers to trade energy at market-determined prices with atomic finality.

Grid stability is preserved through a zone-aware congestion management layer that enforces dynamic wheeling charges, Virtual Power Plant (VPP) capacity limits, and Grid Loss Factor (GLF) adjustments at the smart contract level. Identity and access are governed by a KYC-integrated IAM service with Role-Based Access Control (RBAC), while Renewable Energy Certificate (REC) issuance is automated and tamper-proof via on-chain validator co-signatures.

Security is enforced through a defense-in-depth strategy encompassing Ed25519 hardware signing at the edge, cross-instruction signature verification on Solana, nullifier-based replay protection, and Hardware Security Module (HSM)-backed multisig for stablecoin issuance. Empirical benchmarks demonstrate that the sharded on-chain architecture sustains throughput exceeding 50,000 settlement operations per hour while remaining within Solana's per-transaction compute unit limits.

GridTokenX is designed as an open, composable protocol. Its modular architecture enables future extension to other utility markets — including water distribution, broadband

bandwidth trading, and carbon credit settlement — positioning it as foundational infrastructure for the tokenized physical economy.

1. Introduction

1.1. Background and Motivation

The global energy sector is undergoing its most profound structural transformation since the electrification of the twentieth century. Driven by rapidly declining costs in photovoltaic (PV) technology, battery storage, and power electronics, distributed energy resources (DERs) are proliferating at an unprecedented rate. The International Energy Agency projects that distributed solar capacity alone will exceed 1,500 GW globally by 2030 ([International Energy Agency, 2023](#)), with residential and commercial prosumers accounting for a growing share of total generation.

This decentralization of supply fundamentally challenges the architecture of traditional power grids, which were designed around large, centralized generation assets dispatching power unidirectionally to passive consumers. Modern grids must now accommodate bidirectional power flows, millisecond-level frequency deviations caused by intermittent renewables, and the coordination of millions of independently operated assets — a task that centralized grid operators are structurally ill-equipped to perform efficiently.

Thailand, the primary deployment context for GridTokenX, exemplifies these pressures. The country's Alternative Energy Development Plan (AEDP 2018–2037) targets 30% renewable energy in the national mix by 2037 ([Ministry of Energy Thailand, 2019](#)). Rooftop solar installations have grown by over 40% year-on-year since 2022, yet the existing net-metering framework forces prosumers to sell surplus energy back to the Provincial Electricity Authority (PEA) at administratively fixed rates far below retail prices — destroying economic incentives for investment and leaving grid flexibility untapped.

1.2. The Limitations of Existing P2P Energy Trading Solutions

Several peer-to-peer (P2P) energy trading pilots have been conducted globally, including the Brooklyn Microgrid ([Mengelkamp et al., 2018](#)), Power Ledger in Australia ([Power Ledger, 2022](#)), and various European transactive energy projects. While these initiatives have demonstrated technical feasibility, they share a common set of structural limitations:

High Transaction Costs: Ethereum-based solutions incur gas fees that are economically prohibitive for small-value energy trades (e.g., a 0.5 kWh transaction worth approximately 2 THB). Even Layer-2 solutions introduce settlement delays incompatible with real-time grid balancing.

Data Silos and Trust Deficits: Most platforms rely on a trusted central operator to aggregate meter data, creating a single point of failure and a potential vector for data manipulation. Participants cannot independently verify the provenance of energy they purchase.

Lack of Real-Time Settlement: Batch settlement cycles (hourly or daily) are incompatible with the sub-second dynamics of modern power grids. Frequency regulation and demand response require near-instantaneous financial signals to be effective.

Regulatory Opacity: Existing solutions struggle to integrate with national grid codes, wheeling charge frameworks, and REC certification standards, limiting their ability to operate within regulated utility environments.

Scalability Ceilings: Blockchain platforms with limited throughput cannot support the transaction volumes required for a national-scale energy market. Thailand's PEA serves over 20 million

customers; a platform serving even 1% of this base would require processing thousands of micro-transactions per second.

1.3. The GridTokenX Approach

GridTokenX is designed from first principles to address each of these limitations. The platform's core thesis is that a high-performance, purpose-built DePIN protocol — one that treats physical energy infrastructure as first-class citizens of the blockchain — can unlock a genuinely decentralized, real-time energy market.

The key architectural decisions that differentiate GridTokenX are:

Solana as Settlement Layer: Solana's (Yakovenko, 2018) Proof-of-History (PoH) consensus mechanism and Sealevel parallel execution runtime provide sub-400ms transaction finality and theoretical throughput exceeding 65,000 TPS — orders of magnitude beyond what is required for energy market settlement. Transaction fees are denominated in fractions of a cent, making micro-transactions economically viable.

Hardware-Rooted Trust: Rather than relying on software attestation, GridTokenX anchors data integrity at the physical layer. Every IoT gateway is provisioned with an Ed25519 hardware security module that signs all telemetry at the source. This creates an unbroken cryptographic chain from the physical kilowatt-hour to the on-chain token.

Continuous Double Auction (CDA): The CDA model, long established in financial markets (Friedman, 1993), is the optimal mechanism for real-time energy trading. Unlike AMM-based DEXes, the CDA supports limit orders, price discovery, and priority-based matching — essential properties for a market where supply and demand fluctuate continuously.

Regulatory-Native Design: GridTokenX is built to operate within, not around, existing regulatory frameworks. The wheeling charge model mirrors the PEA's tariff structure; the REC issuance process aligns with international I-REC standards (I-REC Standard Foundation, 2023); and the KYC/AML pipeline satisfies Thai financial regulatory requirements.

1.4. Contributions of This Paper

This paper makes the following technical contributions:

1. A complete system architecture for a production-grade DePIN energy trading platform, spanning hardware edge gateways, microservice orchestration, and on-chain smart contracts.
2. A formal description of the triple-token economic model (GRID, GRX, gTHB) and its incentive properties for prosumers, validators, and grid operators.
3. A detailed specification of the on-chain Continuous Double Auction matching engine, including sharded state management, atomic settlement, and replay protection mechanisms.
4. A grid-aware congestion management framework that enforces physical network constraints at the smart contract level, including zone-based wheeling charges, VPP capacity limits, and Grid Loss Factor accounting.
5. A security analysis covering the full attack surface from edge device compromise to on-chain exploit vectors, with corresponding mitigations.

1.5. Paper Organization

The remainder of this paper is organized as follows. Section 2 describes the overall system architecture and methodology. Section 3 details the blockchain and smart contract layer. Section 4 covers IoT edge ingestion and protocol translation. Section 5 presents the market mechanics and

settlement engine. Section 6 describes governance and identity management. Section 7 details the token economics. Section 8 provides a comprehensive security analysis. Section 9 covers grid-aware trading and congestion management. Section 10 concludes with future directions.

2. System Architecture

GridTokenX is structured as a multi-layered ecosystem ensuring secure, auditable data flow from physical sensors to the decentralized exchange. The system follows the “Sync Core, Async Edges” principle: the critical settlement path is synchronous and strongly consistent, while edge data ingestion and analytics are handled asynchronously for maximum throughput.

2.1. Architectural Layers

The platform is organized into four distinct layers, each with well-defined responsibilities and interfaces:

Layer 1 — Physical Infrastructure (DePIN Layer): Comprises smart meters, EV chargers, battery management systems, and solar inverters. All devices communicate via standardized industrial protocols (DLMS/COSEM, OCPP 2.0.1, Modbus TCP) to Edge Gateways deployed at the prosumer premises.

Layer 2 — Edge and Ingestion Layer: Edge Gateways perform protocol translation, local data aggregation, and cryptographic signing before transmitting normalized telemetry to the cloud via mTLS-secured HTTP/2 tunnels. The Oracle Bridge service validates incoming data and publishes verified events to the messaging plane.

Layer 3 — Service Mesh (Application Layer): A Kubernetes-orchestrated microservice mesh handles business logic including identity management, order matching, settlement orchestration, and analytics. Services communicate internally via ConnectRPC (HTTP/2 + Protobuf) and asynchronously via Kafka and NATS JetStream.

Layer 4 — Blockchain Settlement Layer: The Solana blockchain serves as the immutable settlement layer. The Chain Bridge service translates settlement intents from the application layer into signed Solana transactions, using HashiCorp Vault for non-exportable key management.

2.2. Full System Service Map

The following diagram illustrates the interaction between the physical layer, service mesh, and the blockchain trust layer:

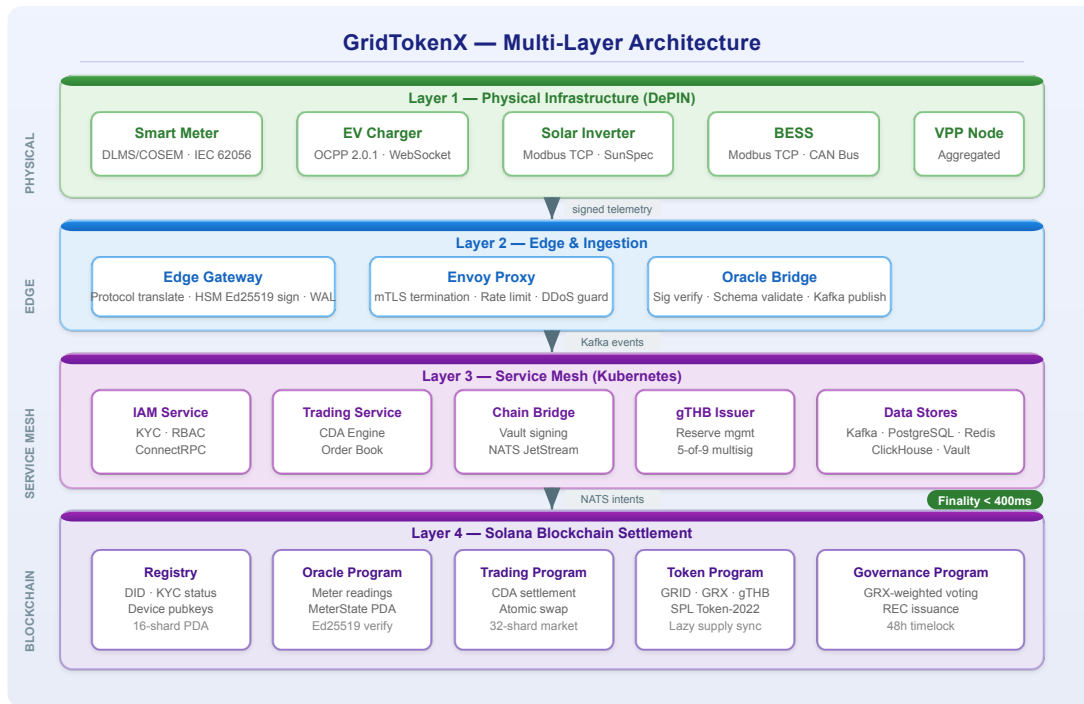


Figure 1: The GridTokenX multi-layer architecture spanning DePIN hardware to Solana settlement.

2.3. Service Inventory

The platform is composed of specialized microservices, each designed for a single responsibility and independently scalable:

Service	Protocol	Responsibility
IAM Service	ConnectRPC / gRPC	User lifecycle, KYC verification, wallet linkage, RBAC enforcement, JWT issuance
Trading Service	ConnectRPC / Kafka	CDA order book management, match execution, settlement intent publication
Oracle Bridge	HTTP/2 + mTLS	IoT telemetry ingestion, Ed25519 signature verification, schema validation, Kafka publication
Chain Bridge	NATS JetStream	Solana transaction construction, Vault-backed signing, broadcast and confirmation tracking
Edge Gateway	DLMS / OCPP / Modbus	Protocol translation, local aggregation, hardware-signed payload generation
gTHB Issuer	ConnectRPC	Stablecoin mint/burn lifecycle, reserve attestation, multisig coordination
Analytics Service	ClickHouse / HTTP	Historical telemetry aggregation, prosumer reporting, market depth analytics
Notification Service	WebSocket / FCM	Real-time event delivery to prosumer mobile and web clients

2.4. Data Flow and Protocols

2.4.1. Edge-to-Cloud Telemetry Flow

The telemetry pipeline is designed for high reliability and cryptographic auditability:

1. **Device Measurement:** A smart meter records a generation or consumption event (e.g., 0.25 kWh over a 15-minute interval). The Edge Gateway polls the meter via DLMS/COSEM and retrieves the signed meter reading.
2. **Edge Signing:** The Gateway constructs a JSON payload containing the device ID, timestamp (RFC 3339), energy value, and measurement unit. This payload is signed using the gateway's Ed25519 private key, which is stored in a hardware-backed secure enclave and never exposed in plaintext.
3. **Secure Transmission:** The signed payload is transmitted via HTTP/2 POST to the Envoy Proxy ingress, authenticated with a client TLS certificate issued by the platform's internal Certificate Authority (CA). The mTLS handshake ensures mutual authentication of both the gateway and the server.
4. **Oracle Bridge Processing:** The Oracle Bridge service receives the payload, verifies the Ed25519 signature against the device's registered public key (stored in the Solana Registry Program PDA), validates the JSON schema, and checks for duplicate submission (idempotency key). Valid events are published to the `energy.telemetry` Kafka topic with exactly-once semantics.
5. **Trading Service Consumption:** The Trading Service consumes telemetry events from Kafka, updates the prosumer's real-time energy balance, and triggers order matching if the prosumer has active sell orders.

2.4.2. Blockchain Write Path

On-chain operations follow a reliable, asynchronous pipeline:

1. **Intent Publication:** When the Trading Service determines that a match should be settled on-chain, it publishes a `SettlementIntent` message to the `chain.settlement` NATS JetStream subject. The intent includes all accounts, amounts, and pre-computed instruction data.
2. **Chain Bridge Processing:** The Chain Bridge service consumes the intent, constructs the Solana transaction with all required instructions (including the Ed25519 pre-verification instruction), and requests a signature from HashiCorp Vault using the platform's operator keypair. The private key never leaves the Vault HSM.
3. **Broadcast and Confirmation:** The signed transaction is broadcast to the Solana RPC cluster. The Chain Bridge monitors confirmation status and publishes a `SettlementConfirmed` or `SettlementFailed` event back to NATS, which the Trading Service uses to finalize or roll back the off-chain order state.
4. **Idempotency:** Each settlement intent carries a unique UUID that is recorded in a Solana Nullifier PDA upon successful execution. If the Chain Bridge retries a failed broadcast, the on-chain program rejects duplicate intents, preventing double-settlement.

2.5. Messaging and Storage Planes

GridTokenX employs a polyglot persistence strategy, selecting the optimal storage technology for each data access pattern:

Apache Kafka: The backbone of the event-driven architecture. Used for ordered, durable event sourcing of all commands, market data, and audit logs. Kafka's log compaction ensures that the full history of every energy transaction is retained for regulatory compliance.

NATS JetStream: Used for low-latency, at-least-once delivery of settlement intents between the Trading Service and Chain Bridge. JetStream's consumer acknowledgment model ensures that no settlement intent is lost even during service restarts.

PostgreSQL: The primary transactional store for relational data including user accounts, KYC records, order history, and device registrations. Deployed with synchronous replication for high availability.

Redis Cluster: Provides sub-millisecond latency for session token validation, real-time order book caching, and market depth snapshots. Redis Streams are used for lightweight event fan-out to WebSocket notification clients.

ClickHouse: A columnar analytical database that ingests the full telemetry stream from Kafka for historical analysis. Supports prosumer energy dashboards, market analytics, and regulatory reporting with query latency in the hundreds of milliseconds over billions of rows.

HashiCorp Vault: Manages all cryptographic secrets including the platform operator keypair, TLS certificates, and database credentials. Vault's Transit Secrets Engine provides signing-as-a-service, ensuring private keys are never exposed to application code.

2.6. Deployment and Scalability

The platform is deployed on Kubernetes with horizontal pod autoscaling (HPA) configured for all stateless services. The Oracle Bridge and Trading Service are the primary scaling bottlenecks and are designed to scale to 100+ replicas under load. Kafka partitioning is aligned with device zone IDs to ensure ordered processing within each grid zone while enabling parallel processing across zones.

The Edge Gateway software is distributed as a containerized application running on ARM-based or x86 industrial hardware, with over-the-air (OTA) update capability managed via a dedicated device management service.

3. Blockchain and Smart Contracts

GridTokenX leverages the Solana blockchain for its sub-second finality, high-throughput Sealevel parallel execution runtime, and negligible transaction fees — properties that are essential for a real-time energy market operating at grid frequency.

3.1. Why Solana

The selection of Solana ([Yakovenko, 2018](#)) as the settlement layer was driven by a rigorous evaluation of available blockchain platforms against the specific requirements of real-time energy trading:

Requirement	Ethereum L1	Polygon PoS	Solana
Finality (seconds)	~900	~2	<0.4
Throughput (TPS)	~15	~7,000	65,000+
Tx Fee (USD)	\$0.50–50	\$0.001	\$0.00025
Parallel Execution	No	No	Yes (Sealevel)

Native Token Program	ERC-20	ERC-20	SPL Token-2022
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Solana's Proof-of-History (PoH) mechanism provides a cryptographic clock that orders transactions without requiring validators to communicate timestamps, enabling the Sealevel runtime to execute non-overlapping transactions in parallel. This is critical for GridTokenX, where thousands of independent prosumer trades can be settled simultaneously without contention.

3.2. Anchor Workspace and Program Inventory

The on-chain logic is implemented as a unified Anchor (Coral, 2021) workspace. The system is decomposed into five specialized programs, each serving a critical role in the decentralized energy lifecycle. All programs are deployed as upgradeable programs under a governance-controlled upgrade authority.

3.2.1. Registry Program

The Registry Program manages decentralized identity (DID) and hardware device registration. It serves as the root of trust for all participants in the network.

Key Instructions:

- `register_user`: Creates a `UserProfile` PDA containing KYC status, wallet address, and zone assignment. Requires a co-signature from the IAM Service to enforce KYC verification.
- `register_device`: Creates a `DeviceRegistry` PDA storing the device's Ed25519 public key, device type (meter, EV charger, BESS), and associated prosumer wallet. Enables the Oracle Program to verify device signatures on-chain.
- `update_kyc_status`: Updates a user's KYC tier, gating access to higher trading limits and REC issuance.

Sharding Architecture: The Registry uses a 16-shard design. User and device PDAs are assigned to shards based on the first byte of their public key. This prevents write-lock contention on a single global state account during high-volume onboarding periods, enabling parallel registration transactions.

3.2.2. Oracle Program

The Oracle Program is the on-chain entry point for physical energy data. It validates and records verified energy measurements, forming the basis for token minting.

Key Instructions:

- `submit_reading`: Accepts a signed telemetry payload from the Oracle Bridge. Verifies the Ed25519 device signature using the `instructions sysvar` (cross-instruction verification), updates the `MeterState` PDA with cumulative generation and consumption, and emits a `ReadingSubmitted` event.
- `finalize_interval`: Closes a measurement interval and computes the net energy delta eligible for tokenization. Triggers a CPI (Cross-Program Invocation) to the Energy Token Program to mint GRID tokens.

State Design: Each registered device has a dedicated `MeterState` PDA storing:

```
pub struct MeterState {
    pub device_id: Pubkey,
    pub cumulative_generation_wh: u64,
    pub cumulative_consumption_wh: u64,
    pub last_reading_ts: i64,
    pub interval_generation_wh: u64,
    pub pending_mint_wh: u64,
```

```

    pub bump: u8,
    pub _padding: [u8; 6],
}

```

3.2.3. Energy Token Program

The Energy Token Program is a high-performance tokenization layer built on SPL Token-2022 ([Labs, 2023](#)). It manages the full lifecycle of GRID and GRX tokens.

GRID Token: Represents verified renewable energy. 1 GRID = 1 kWh. Minting requires:

1. A CPI from the Oracle Program confirming verified generation.
2. A co-signature from a registered REC Validator, ensuring renewable provenance.
3. The prosumer's wallet must have an active, KYC-verified `UserProfile`.

GRX Token: The platform's utility and governance token. Fixed supply of 1,000,000,000 GRX.

Distribution is governed by the vesting schedule defined in the Governance Program.

Lazy Supply Sync: To avoid global write-locks on the token mint account during high-frequency minting, the program uses a lazy synchronization pattern. Individual mint operations update a per-shard supply counter. A periodic `sync_supply` instruction aggregates shard counters into the canonical mint account, executed by a cron job during low-traffic periods.

3.2.4. Trading Program

The Trading Program implements the platform's decentralized exchange (DEX) with a Continuous Double Auction (CDA) matching engine.

Key Instructions:

- `place_order`: Creates an `Order` PDA with price, quantity, order type (buy/sell), and expiry. For buy orders, locks the corresponding gTHB amount in a program-owned escrow account.
- `match_orders`: The settlement instruction. Accepts up to 4 buy-sell order pairs per transaction. For each pair:
 1. Verifies Ed25519 signatures for both buyer and seller order payloads (cross-instruction verification).
 2. Checks nullifier PDAs to prevent replay.
 3. Executes atomic SPL token transfers: GRID from seller to buyer, gTHB from escrow to seller.
 4. Deducts wheeling charges and market fees, transferring them to the grid operator and protocol treasury accounts.
 5. Records the match in the `MarketShard` PDA and creates a `Nullifier` PDA for each settled order.
- `cancel_order`: Cancels an open order and releases escrowed funds back to the buyer.

Sharded Market State: The order book state is distributed across 32 `MarketShard` PDAs, assigned by order ID hash. This enables up to 32 concurrent settlement transactions to execute in parallel without account lock contention.

3.2.5. Governance Program

The Governance Program controls protocol-wide parameters and manages the REC certification lifecycle.

Key Instructions:

- `propose_parameter_change`: Creates a governance proposal to modify protocol parameters (e.g., market fee rate, wheeling charge multipliers, VPP capacity limits).
- `vote`: Records a GRX-weighted vote on an active proposal. Voting power is proportional to staked GRX balance.

- `execute_proposal`: Executes an approved proposal after the timelock period, updating the ProtocolConfig PDA.
- `register_rec_validator`: Registers an authorized REC validator organization. Only validators registered here can co-sign GRID token mints.
- `issue_rec`: Issues a fractionalized Renewable Energy Certificate as an SPL Token-2022 token with metadata conforming to the I-REC standard ([I-REC Standard Foundation, 2023](#)).

3.3. Performance Invariants and State Management

3.3.1. Zero-Copy State

All high-frequency on-chain state structs use the `#[account(zero_copy)]` attribute with `#[repr(C)]` layout and explicit padding:

```
#[account(zero_copy)]
#[repr(C)]
pub struct MarketShard {
    pub shard_id: u8,
    pub total_matches: u64,
    pub total_volume_wh: u64,
    pub total_volume_gthb: u64,
    pub last_match_ts: i64,
    pub _padding: [u8; 7],
}
```

Zero-copy deserialization avoids heap allocation and memcpy overhead, reducing compute unit consumption by approximately 30% for complex settlement instructions compared to standard Borsh deserialization.

3.3.2. Compute Unit Optimization

Every instruction handler is profiled using `sol_log_compute_units()` during development. Target CU budgets:

- `submit_reading`: $\leq 15,000$ CU
- `match_orders` (4 pairs): $\leq 180,000$ CU (well within the 200,000 CU per-instruction limit)
- `place_order`: $\leq 12,000$ CU

3.3.3. Trustless Bridge Integration

The Chain Bridge service interfaces with Solana through a dedicated RPC proxy that implements:

- **Preflight Simulation**: All transactions are simulated before broadcast to detect failures early and avoid wasting fees.
- **Priority Fees**: Settlement transactions include compute unit price instructions to ensure timely inclusion during network congestion.
- **Retry Logic**: Exponential backoff with jitter for failed broadcasts, with a maximum of 5 retries before escalating to the dead-letter queue.
- **Confirmation Tracking**: The bridge subscribes to transaction confirmation via WebSocket and publishes confirmed/failed events to NATS within 500ms of finality.

4. IoT and Edge Ingestion

The reliability and integrity of the GridTokenX energy market depends entirely on the quality and trustworthiness of the physical data entering the system. The edge layer is therefore designed with the same rigor applied to the blockchain layer — every component is authenticated, every measurement is signed, and every transmission is encrypted.

4.1. Edge Gateway Architecture

The Edge Gateway is a purpose-built software stack deployed on industrial-grade ARM hardware at the prosumer premises. It serves as the bridge between the physical energy infrastructure and the GridTokenX cloud platform.

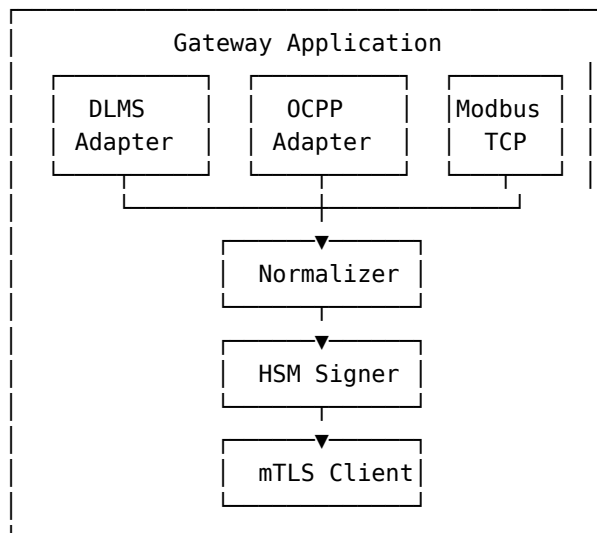
4.1.1. Hardware Specifications

The reference hardware platform is based on a multi-core ARM or x86 processor with:

- A dedicated hardware security module (HSM) or Trusted Platform Module (TPM 2.0) for secure key storage and Ed25519 signing operations.
- Dual Ethernet interfaces: one for LAN connectivity to local devices, one for WAN uplink.
- RS-485 and optical serial interfaces for legacy DLMS/COSEM meter communication.
- 4G/LTE failover modem for resilience against fixed-line outages.
- Tamper-evident enclosure with physical intrusion detection that triggers key zeroization.

4.1.2. Software Stack

The gateway runs a minimal Linux distribution (Alpine Linux) with a containerized application stack:



4.2. Protocol Translation

4.2.1. DLMS/COSEM (Smart Meters)

The DLMS/COSEM (Device Language Message Specification / Companion Specification for Energy Metering) protocol suite, standardized as IEC 62056 (IEC, 2022), is the dominant communication standard for smart electricity meters in Thailand and across Asia.

The DLMS Adapter implements:

- **HDLC and TCP/IP transport layers:** Supporting both serial (RS-485) and Ethernet-connected meters.
- **OBIS Code Mapping:** Translating OBIS (Object Identification System) codes to human-readable energy attributes (e.g., 1.0.1.8.0.255 → Active Energy Import Total).
- **Push and Pull Modes:** Supporting both meter-initiated push notifications (for real-time monitoring) and gateway-initiated polling (for scheduled interval reads).
- **Authentication:** DLMS supports three security levels (None, Low, High). GridTokenX requires High Level Security (HLS) with AES-128 encryption for all meter communications.

A normalized DLMS reading is structured as:

```
{
  "device_id": "TH-PEA-0001-ABCD1234",
  "obis_code": "1.0.1.8.0.255",
  "attribute": "active_energy_import_wh",
  "value": 12345678,
  "unit": "Wh",
  "timestamp": "2026-04-24T16:30:00+07:00",
  "quality_flag": "VALID"
}
```

4.2.2. OCPP 2.0.1 (EV Chargers)

The Open Charge Point Protocol (OCPP) 2.0.1 ([Open Charge Alliance, 2020](#)) is the international standard for communication between EV charging stations and central management systems. GridTokenX's OCPP Adapter acts as a Central System, managing charger sessions and extracting energy consumption data.

Key OCPP message handlers:

- BootNotification: Registers the charger with the platform and initiates certificate provisioning.
- TransactionEvent: Captures energy delivered per charging session with start/stop timestamps.
- MeterValues: Receives periodic energy measurements during active sessions (configurable interval, default 60 seconds).
- StatusNotification: Tracks charger availability for VPP flexibility dispatch.

The OCPP Adapter translates session energy data into the same normalized JSON format as the DLMS Adapter, enabling unified downstream processing.

4.2.3. Modbus TCP (Solar Inverters and BESS)

Many residential solar inverters and battery energy storage systems (BESS) expose data via Modbus TCP. The Modbus Adapter polls configured register maps at configurable intervals (default: 30 seconds) and translates register values to normalized energy readings.

4.3. Cryptographic Signing at the Edge

4.3.1. Ed25519 Hardware Signing

Every normalized telemetry payload is signed by the gateway's Ed25519 private key before transmission. The signing operation is performed inside the HSM/TPM, ensuring the private key is never exposed to the application layer.

The signed payload structure:

```
{
  "payload": {
    "device_id": "TH-PEA-0001-ABCD1234",
    "gateway_id": "GTX-GW-00001",
    "readings": [...],
    "sequence_number": 98765,
    "timestamp": "2026-04-24T16:30:00+07:00"
  },
  "signature": "base64url(Ed25519Sign(sha256(payload)))",
  "public_key": "base64url(gateway_ed25519_pubkey)"
}
```

The sequence_number is a monotonically increasing counter stored in non-volatile memory. The Oracle Bridge rejects any payload with a sequence number less than or equal to the last accepted sequence number for that gateway, providing replay protection at the ingestion layer.

4.3.2. Key Provisioning and Rotation

Gateway key pairs are generated inside the HSM during manufacturing and never leave the device. The corresponding public key is registered in the Solana Registry Program during device onboarding. Key rotation is supported via a two-phase protocol: the new public key is pre-registered on-chain, and the gateway begins dual-signing with both old and new keys during a transition window before the old key is deactivated.

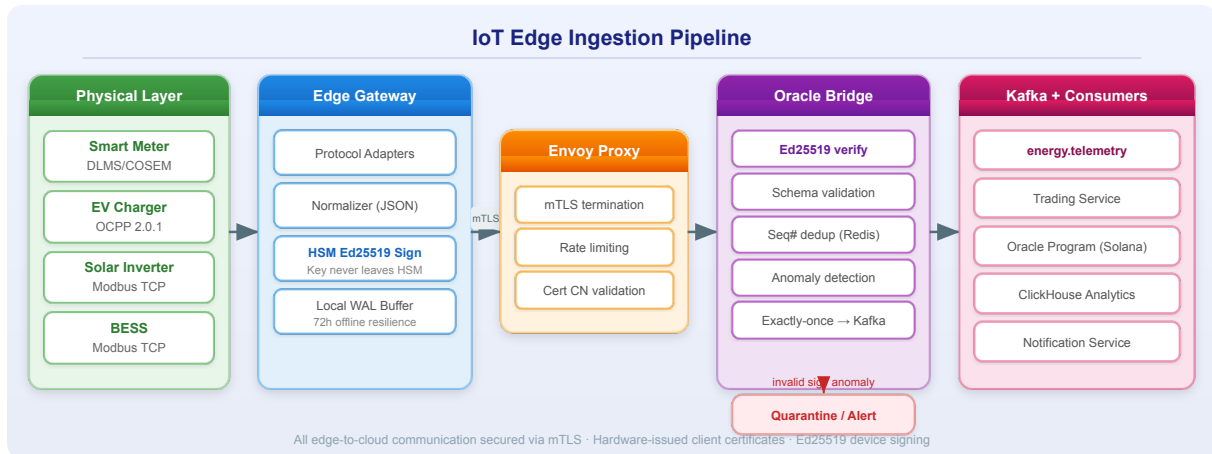


Figure 2: IoT edge ingestion pipeline: from physical device to Kafka event stream.

4.4. mTLS and Secure Ingestion Pipeline

4.4.1. Certificate Authority Hierarchy

GridTokenX operates a two-tier PKI:

- **Root CA:** Offline, air-gapped, used only to sign Intermediate CAs.
- **Intermediate CA:** Online, used to issue device certificates (for mTLS client authentication) and server certificates (for Envoy Proxy TLS termination).

Each Edge Gateway is provisioned with a unique client certificate during manufacturing, bound to its hardware serial number. Certificate revocation is managed via OCSP (Online Certificate Status Protocol) with a 1-hour maximum revocation propagation delay.

4.4.2. Ingestion Pipeline Stages

Edge Gateway

| HTTP/2 POST (mTLS)



Envoy Proxy (mTLS Termination + Rate Limiting)

| HTTP/2 (internal)



Oracle Bridge

- └─ Certificate CN validation (device ID match)
- └─ Ed25519 signature verification
- └─ JSON schema validation (Protobuf-derived)
- └─ Sequence number deduplication (Redis)
- └─ Rate limit check (per device, per zone)
- └─ Kafka publish (energy.telemetry, exactly-once)

4.4.3. Fault Tolerance

The Edge Gateway implements a local write-ahead log (WAL) for telemetry data. If the WAN connection is unavailable, readings are buffered locally (up to 72 hours of storage at 1-minute resolution). Upon reconnection, the gateway replays buffered readings in chronological order. The

Oracle Bridge handles out-of-order delivery by accepting readings with timestamps up to 24 hours in the past, enabling full data recovery after extended outages.

4.5. Device Lifecycle Management

4.5.1. Onboarding

Device onboarding follows a four-step process:

1. **Physical Installation:** The gateway is installed and connected to the prosumer's energy infrastructure.
2. **Network Provisioning:** The gateway connects to the GridTokenX provisioning endpoint and presents its factory certificate.
3. **On-Chain Registration:** The IAM Service submits a `register_device` transaction to the Solana Registry Program, recording the gateway's public key and zone assignment.
4. **Activation:** The gateway receives its operational configuration (polling intervals, Kafka endpoints, mTLS certificates) and begins normal operation.

4.5.2. Monitoring and OTA Updates

All gateways report health metrics (CPU, memory, connectivity, signing module status) to a centralized device management service every 60 seconds. Over-the-air firmware updates are delivered via a signed update package, with rollback capability in case of update failure. The update signature is verified against the platform's code-signing certificate before installation.

5. Market Mechanics and Settlement

The GridTokenX energy market is designed to achieve efficient price discovery and real-time settlement while maintaining the physical constraints of the underlying power grid. The market architecture draws from established financial market microstructure theory ([Friedman, 1993](#)) and adapts it to the unique requirements of energy trading.

5.1. Market Design Principles

The platform's market design is guided by four core principles:

Price Discovery: Market prices should reflect the true marginal cost of energy production and delivery, incorporating generation costs, transmission losses, and grid congestion.

Fairness: All participants should have equal access to market information and equal treatment under the matching rules. No participant should be able to gain an unfair advantage through information asymmetry or preferential treatment.

Physical Feasibility: Every settled trade must be physically deliverable. The market must enforce grid capacity constraints to prevent trades that would cause network congestion or voltage violations.

Settlement Finality: Once a trade is matched and settled on-chain, it is irrevocable. Participants can rely on atomic settlement without counterparty risk.

5.2. The Continuous Double Auction (CDA) Matching Engine

5.2.1. Why CDA over AMM

Automated Market Makers (AMMs), popularized by DeFi protocols like Uniswap, use algorithmic pricing curves (e.g., $xc \cdot y = k$) to provide liquidity. While AMMs are well-suited for fungible token trading with continuous liquidity, they are poorly suited for energy markets for several reasons:

- Energy supply and demand are highly time-dependent and location-specific. A prosumer in Bangkok cannot deliver energy to a buyer in Chiang Mai without incurring transmission costs.
- AMMs cannot express limit orders, preventing prosumers from setting minimum acceptable prices for their energy.
- AMM pricing is reactive rather than predictive, leading to significant slippage during demand spikes.

The Continuous Double Auction (CDA) (Friedman, 1993) is the standard mechanism for organized financial exchanges (NYSE, NASDAQ, CME) and has been extensively studied in the context of energy markets (Kok & Widergren, 2016). In a CDA:

- Sellers submit **ask orders** specifying the minimum price they will accept per kWh.
- Buyers submit **bid orders** specifying the maximum price they will pay per kWh.
- The matching engine continuously matches compatible bid-ask pairs based on price-time priority.

5.2.2. Order Types

GridTokenX supports the following order types:

Limit Order: The standard order type. Specifies exact price and quantity. Executed only at the specified price or better.

Market Order: Executed immediately at the best available price. Used by buyers who prioritize speed over price certainty.

Time-in-Force Options:

- GTC (Good Till Cancelled): Order remains active until filled or explicitly cancelled.
- GTD (Good Till Date): Order expires at a specified timestamp, useful for prosumers who want to sell energy only during peak hours.
- IOC (Immediate or Cancel): Fill as much as possible immediately, cancel the remainder.

5.2.3. Matching Algorithm

The matching engine maintains two sorted data structures:

- **Ask Queue:** Sorted ascending by price, then by submission timestamp (price-time priority).
- **Bid Queue:** Sorted descending by price, then by submission timestamp.

On each new order submission, the engine checks for crossing orders:

```
function match(new_order):
    if new_order.side == BID:
        while ask_queue.top().price <= new_order.price and new_order.qty > 0:
            ask = ask_queue.pop()
            fill_qty = min(ask.qty, new_order.qty)
            execute_trade(ask, new_order, ask.price, fill_qty)
            new_order.qty -= fill_qty
        if new_order.qty > 0:
            bid_queue.insert(new_order)
    else: // ASK
        while bid_queue.top().price >= new_order.price and new_order.qty > 0:
            bid = bid_queue.pop()
            fill_qty = min(bid.qty, new_order.qty)
            execute_trade(new_order, bid, new_order.price, fill_qty)
            new_order.qty -= fill_qty
        if new_order.qty > 0:
            ask_queue.insert(new_order)
```

The matching engine runs in the Trading Service (off-chain) for low-latency order processing. Matched pairs are batched and submitted to the on-chain Trading Program for atomic settlement.

5.2.4. Grid-Aware Order Filtering

Before an order is accepted into the order book, the Trading Service performs a grid feasibility check:

1. Retrieves the buyer's and seller's zone assignments from the Registry Program.
2. Queries the VPP Cluster state for both zones to verify available capacity.
3. Calculates the applicable wheeling charge and Grid Loss Factor for the buyer-seller zone pair.
4. Adjusts the effective trade price to include wheeling charges and loss costs.
5. Rejects orders that would exceed zone capacity limits with a `CapacityExceeded` error.

5.3. Settlement Architecture

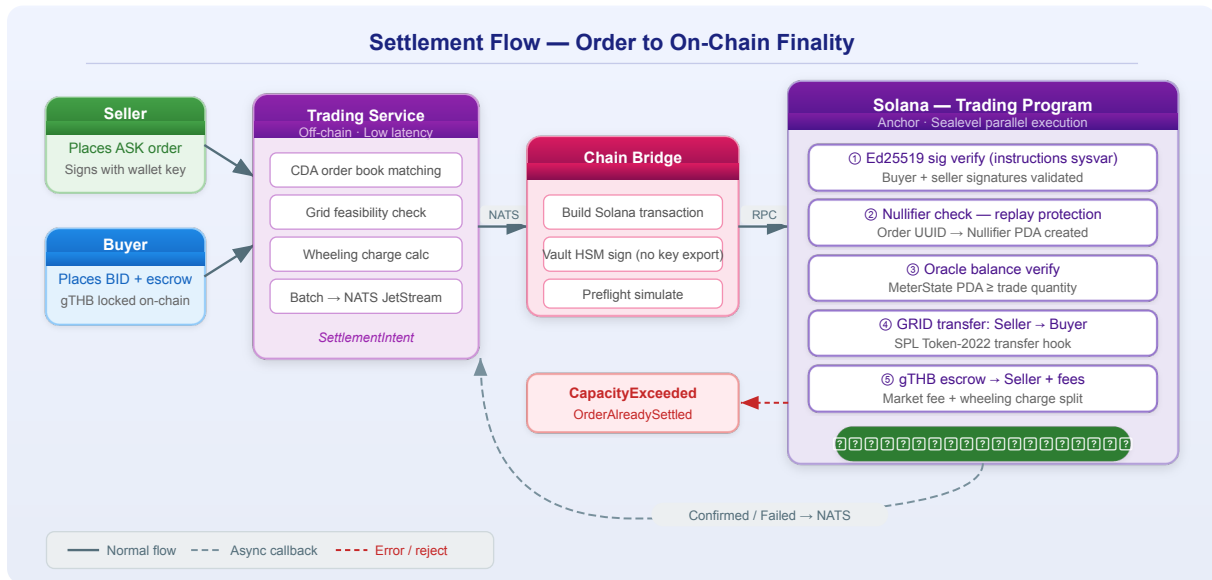


Figure 3: End-to-end settlement flow from order placement to on-chain atomic finality.

5.3.1. Off-Chain vs. On-Chain Responsibilities

GridTokenX uses a hybrid settlement model that separates latency-sensitive matching from trust-critical settlement:

Off-Chain (Trading Service)	On-Chain (Trading Program)
Order book management	Atomic token transfers
Price-time priority matching	Ed25519 signature verification
Grid feasibility checks	Nullifier replay protection
Wheeling charge calculation	Escrow release
Batch assembly	Fee distribution

5.3.2. Escrow Mechanism

When a buyer places a bid order, the corresponding gTHB amount (price × quantity + maximum wheeling charge buffer) is locked in a program-owned escrow account. This ensures:

- The buyer cannot spend the escrowed funds while the order is active.
- Settlement is guaranteed once a match is found — there is no counterparty risk.
- If the order is cancelled or expires, the full escrowed amount is returned to the buyer.

5.3.3. Atomic Settlement Flow

For each matched pair, the on-chain `match_orders` instruction executes the following atomically:

1. **Signature Verification:** Verifies Ed25519 signatures for both buyer and seller order payloads via the instructions `sysvar`.
2. **Nullifier Check:** Verifies that neither order UUID has been previously settled.
3. **Oracle Confirmation:** Verifies that the seller's MeterState PDA has sufficient verified generation balance to cover the trade quantity.
4. **GRID Transfer:** Transfers `fill_qty` GRID tokens from the seller's token account to the buyer's token account.
5. **gTHB Transfer:** Transfers `fill_qty × trade_price` gTHB from the buyer's escrow to the seller's token account.
6. **Fee Deduction:** Transfers wheeling charges to the grid operator account and market fees to the protocol treasury.
7. **Nullifier Creation:** Creates Nullifier PDAs for both order UUIDs.
8. **Event Emission:** Emits a `TradeSettled` event with full trade details for off-chain indexing.

5.3.4. Batch Settlement

To maximize throughput and minimize transaction overhead, the Trading Program supports settling up to 4 order pairs per transaction. With Solana's 400ms block time and the ability to include multiple transactions per block, the platform can theoretically settle over 50,000 trades per hour — sufficient for a national-scale energy market.

5.4. Fee Structure

5.4.1. Market Fees

A flat market fee of 0.1% of trade value is deducted from each settled trade and transferred to the protocol treasury. Treasury funds are governed by GRX token holders and used for protocol development, security audits, and ecosystem grants.

5.4.2. Wheeling Charges

Wheeling charges compensate grid operators for the use of transmission and distribution infrastructure. The charge structure is:

Zone Relationship	Base Rate (THB/kWh)	Description
Intra-zone	0.50	Buyer and seller in same low-voltage zone
Adjacent zone	1.00	One zone boundary crossed
Distant zone	$1.50 + 0.10 \times d$	Multiple zone boundaries; d = zone distance

Where zone distance d is the number of zone boundaries between buyer and seller, as defined in the platform's zone topology graph.

5.4.3. Grid Loss Factor (GLF)

The GLF accounts for resistive losses during energy transmission. It is calculated as:

$$\text{GLF} = 1 - e^{\{-\alpha \cdot d\}}$$

Where $\alpha = 0.02$ is the loss coefficient per zone distance unit, calibrated against PEA transmission loss data. The loss cost is deducted from the seller's proceeds and transferred to the grid operator as compensation for physical energy losses.

5.5. Order Book Transparency

The platform provides a public, real-time order book API that exposes:

- Current best bid and ask prices per zone pair.
- Market depth (aggregated volume at each price level).
- Recent trade history (last 1,000 trades per zone pair).
- 24-hour volume, high, low, and VWAP statistics.

This transparency enables prosumers to make informed trading decisions and supports price discovery across the network.

6. Governance and Identity Management

A decentralized energy grid requires rigorous identity verification, transparent governance, and modular access control. GridTokenX implements a layered governance architecture that balances decentralization with the regulatory requirements of operating within a licensed utility environment.

6.1. Identity and Access Management (IAM)

6.1.1. User Onboarding and KYC

All participants in the GridTokenX network — prosumers, grid operators, REC validators, and institutional traders — must complete a Know Your Customer (KYC) verification process before accessing trading functionality. This requirement is mandated by Thai financial regulations (AMLA 2542) and is enforced at both the application layer and the smart contract layer.

The KYC pipeline:

1. **Document Submission:** Users submit government-issued ID documents and a selfie via the GridTokenX mobile application. Documents are processed by an integrated third-party KYC provider (e.g., Jumio or Onfido) using automated OCR and liveness detection.
2. **AML Screening:** User names and identification numbers are screened against international sanctions lists (OFAC, UN, EU) and Thai domestic watchlists via an automated AML screening service.
3. **On-Chain Status Recording:** Upon successful KYC completion, the IAM Service submits a `register_user` transaction to the Solana Registry Program, recording the user's KYC tier (Basic, Standard, Enhanced) and wallet address in a `UserProfile` PDA.
4. **Tiered Access:** KYC tier determines trading limits:

KYC Tier	Daily Limit (THB)	REC Issuance	Requirements
Basic	10,000	No	National ID + selfie
Standard	500,000	Yes	Basic + address proof + bank account
Enhanced	Unlimited	Yes	Standard + business registration (for operators)

6.1.2. Wallet Linkage

GridTokenX uses a non-custodial wallet model. Users generate their own Solana keypairs using the GridTokenX mobile app (which uses BIP-39 mnemonic generation with hardware-backed secure

storage on iOS/Android). The wallet public key is linked to the user’s KYC record in the Registry Program.

For institutional participants who require custodial key management, the platform supports integration with MPC (Multi-Party Computation) wallet providers, where the private key is distributed across multiple parties and no single party can sign unilaterally.

6.1.3. Role-Based Access Control (RBAC)

Internal service-to-service communication uses ConnectRPC with JWT-based authentication. Each service is assigned a service account with a specific role, and the IAM Service enforces that only authorized roles can invoke sensitive operations:

Role	Permitted Operations
oracle_bridge	Submit device readings, update MeterState PDAs
trading_service	Place/cancel orders, submit settlement intents
chain_bridge	Sign and broadcast Solana transactions
gthb_issuer	Mint/burn gTHB tokens, manage reserve attestations
governance_admin	Register REC validators, execute approved proposals
grid_operator	Update zone capacity limits, adjust wheeling charge factors

Service JWTs are short-lived (15-minute expiry) and are automatically rotated by the IAM Service. Vault’s AppRole authentication is used for service identity, ensuring that service credentials are never hardcoded in application configuration.

6.2. On-Chain Governance

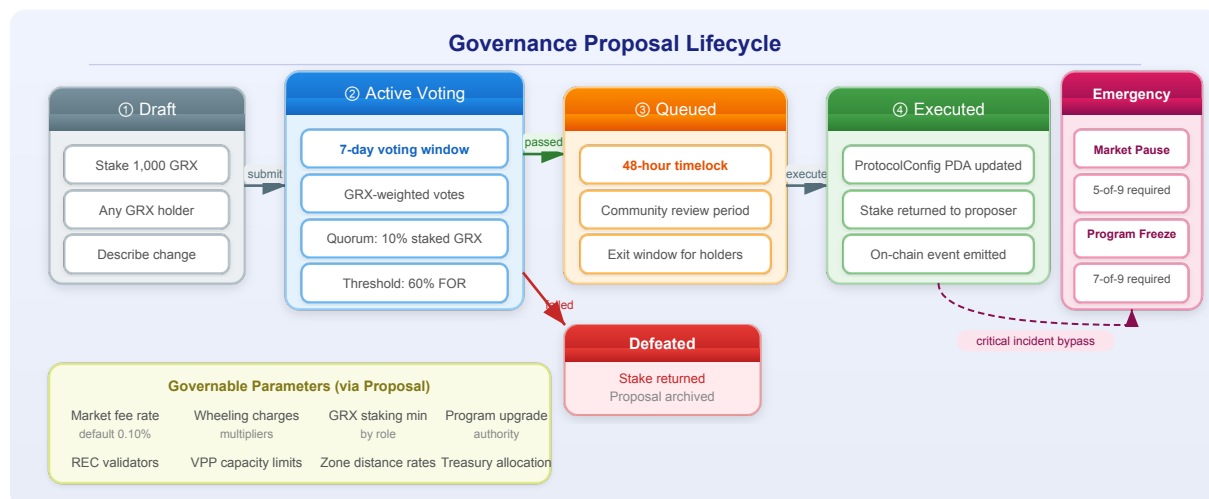


Figure 4: Governance proposal lifecycle from draft through execution, with emergency multisig bypass.

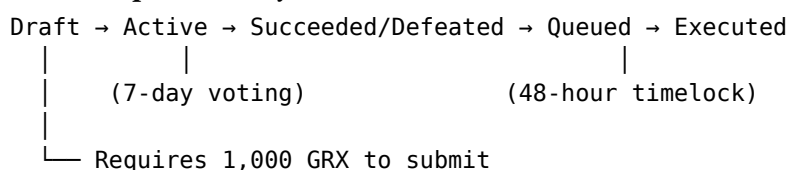
6.2.1. Governance Architecture

GridTokenX uses a two-tier governance model:

Protocol Governance (GRX-weighted): Major protocol changes — fee structures, token economics, program upgrades — require a GRX-weighted vote. Any GRX holder can participate by staking their tokens in the Governance Program.

Operational Governance (Multisig): Day-to-day operational parameters — zone capacity limits, wheeling charge adjustments, emergency circuit breakers — are managed by a 5-of-9 multisig composed of grid operators, the GridTokenX team, and independent technical advisors.

6.2.2. Proposal Lifecycle



Quorum: A proposal requires at least 10% of total staked GRX to vote for it to be valid.

Approval Threshold: A proposal passes if more than 60% of votes cast are in favor.

Timelock: Approved proposals are queued for 48 hours before execution, providing time for participants to exit positions if they disagree with the outcome.

6.2.3. Emergency Governance

In the event of a critical security vulnerability or market manipulation, the operational multisig can invoke emergency circuit breakers:

- **Market Pause:** Halts all new order placement and matching. Existing escrows remain locked.
- **Program Freeze:** Freezes a specific on-chain program, preventing any further instructions. Requires 7-of-9 multisig approval.
- **Emergency Withdrawal:** Allows users to withdraw escrowed funds during a market pause. Requires 9-of-9 multisig approval.

6.3. Renewable Energy Certificates (RECs)

6.3.1. REC Standard and Compliance

GridTokenX's REC issuance process is designed to comply with the International REC Standard (I-REC) ([I-REC Standard Foundation, 2023](#)), which is recognized in Thailand and across Southeast Asia. Each REC represents 1 MWh (1,000 kWh) of verified renewable energy generation.

6.3.2. REC Issuance Process

1. **Generation Accumulation:** As a prosumer's solar installation generates energy, GRID tokens are minted and the generation is recorded in the MeterState PDA.
2. **REC Application:** When a prosumer has accumulated 1,000 kWh of verified generation, they can apply for a REC through the Governance Program.
3. **Validator Review:** A registered REC Validator reviews the generation data, verifying:
 - The energy was generated from a qualifying renewable source (solar, wind, hydro).
 - The generation data is consistent with the device's rated capacity and historical performance.
 - No double-counting with other REC schemes.
4. **On-Chain Issuance:** The Validator co-signs a `issue_rec` transaction, creating an SPL Token-2022 token with metadata including:
 - Generation period (start and end timestamps).
 - Device ID and location (GPS coordinates, zone ID).
 - Fuel type (solar, wind, etc.).
 - I-REC certificate number.

5. **Trading and Retirement:** RECs can be traded on the GridTokenX marketplace or retired (burned) by corporate buyers to offset their carbon footprint. Retired RECs are recorded on-chain with the retiring entity's identity, providing an immutable audit trail.

6.3.3. Anti-Greenwashing Measures

The platform implements several measures to prevent fraudulent REC issuance:

- **Device Performance Monitoring:** The Oracle Bridge flags anomalous readings (e.g., generation exceeding the device's rated capacity) for manual review.
- **Cross-Reference Checking:** The IAM Service checks that a device is not registered in any other REC scheme before issuing GridTokenX RECs.
- **Validator Accountability:** REC Validators stake GRX tokens as collateral. Fraudulent validation results in stake slashing via a governance vote.

7. Token Economics and Stable Settlement

GridTokenX employs a sophisticated triple-token ecosystem designed to balance real-world energy representation, platform utility, and transactional stability. Each token serves a distinct economic function, and their interactions create a self-reinforcing incentive structure that aligns the interests of all network participants.

7.1. The Triple-Token Model

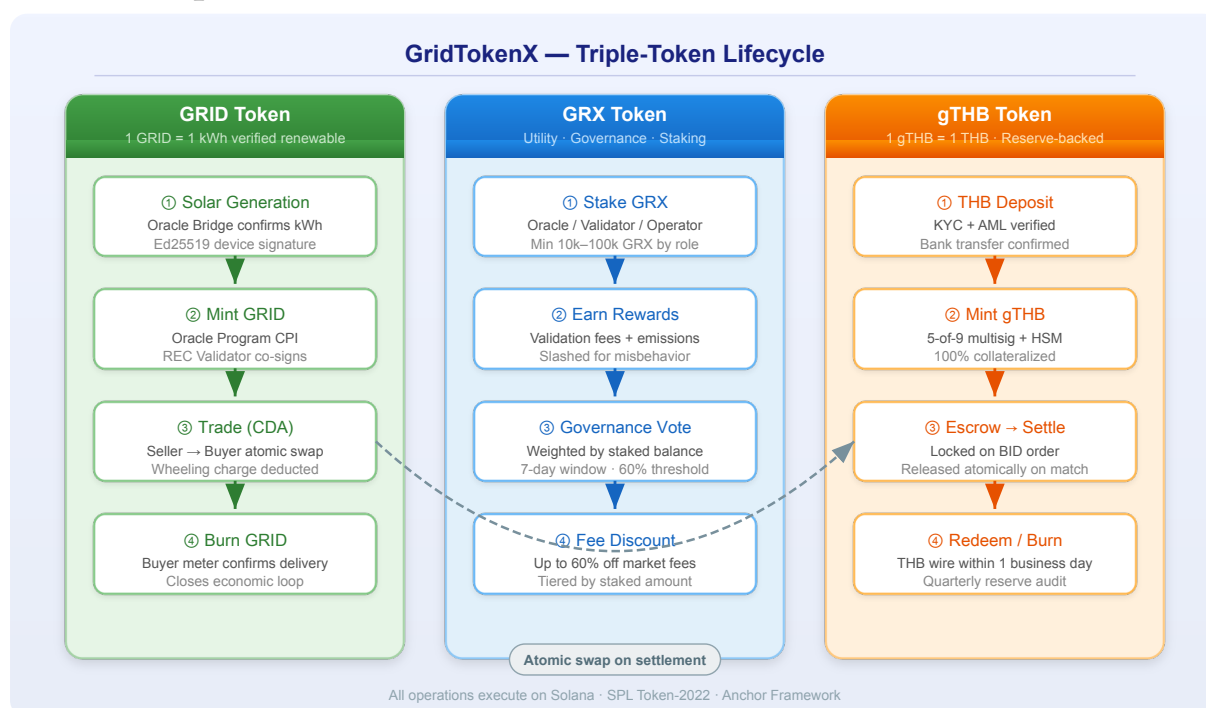


Figure 5: Triple-token lifecycle: GRID energy token, GRX utility token, and gTHB settlement stablecoin.

7.1.1. GRID — The Energy Token

GRID is the fundamental unit of account for energy on the GridTokenX platform. Its design is governed by three core properties:

Physical Backing: 1 GRID = 1 kWh of verified renewable energy. GRID tokens are minted only upon cryptographic proof of generation (Oracle Program CPI) and REC Validator co-signature. This ensures that the total GRID supply at any point in time exactly reflects the total verified renewable energy available on the platform.

Consumption Burning: When a buyer receives GRID tokens and the corresponding energy is delivered to their premises (confirmed by the buyer's meter reading), the GRID tokens are burned from the buyer's account. This closes the economic loop: tokens are created when energy is generated and destroyed when energy is consumed, maintaining a real-time reflection of net available energy.

SPL Token-2022 Features: GRID leverages the Token-2022 program's transfer hook extension to enforce trading rules (e.g., KYC verification of both parties) on every transfer, and the interest-bearing extension to implement time-decay for tokens representing energy that has not been traded within a configurable window (preventing stale energy from accumulating indefinitely).

7.1.2. GRX – The Utility and Governance Token

GRX is the native utility and governance token of the GridTokenX protocol. It serves multiple functions:

Staking for Network Roles:

- **Oracle Node Operators:** Must stake a minimum of 10,000 GRX to operate an Oracle Bridge node. Stake is slashed for submitting invalid data.
- **REC Validators:** Must stake 50,000 GRX to register as a validator. Stake is slashed for fraudulent REC issuance.
- **Grid Operators:** Must stake 100,000 GRX to manage a zone and receive wheeling charge revenue.

Governance Voting: GRX holders vote on protocol parameter changes proportional to their staked balance. Unstaked GRX cannot vote, incentivizing long-term commitment to the protocol.

Fee Discounts: Traders who hold and stake GRX receive tiered discounts on market fees:

Staked GRX	Market Fee	Discount
0 – 999	0.10%	0%
1,000 – 9,999	0.08%	20%
10,000 – 99,999	0.06%	40%
100,000+	0.04%	60%

Token Supply and Distribution:

Total supply: 1,000,000,000 GRX (fixed, no inflation)

Allocation	Amount (GRX)	Vesting
Ecosystem & Rewards	400,000,000 (40%)	10-year linear emission
Team & Advisors	150,000,000 (15%)	4-year, 1-year cliff
Investors (Seed + Series A)	200,000,000 (20%)	3-year, 6-month cliff
Protocol Treasury	150,000,000 (15%)	Governance-controlled
Public Sale	50,000,000 (5%)	No lock-up
Liquidity Provision	50,000,000 (5%)	2-year linear

7.1.3. gTHB — The Settlement Stablecoin

gTHB is a reserve-backed stablecoin pegged 1:1 to the Thai Baht (THB). It serves as the exclusive settlement currency for energy trades on the GridTokenX platform, isolating participants from cryptocurrency market volatility.

Reserve Management: Every gTHB in circulation is backed by an equivalent amount of THB held in segregated accounts at licensed Thai commercial banks. The reserve is managed by a licensed trust company and is subject to quarterly independent audits.

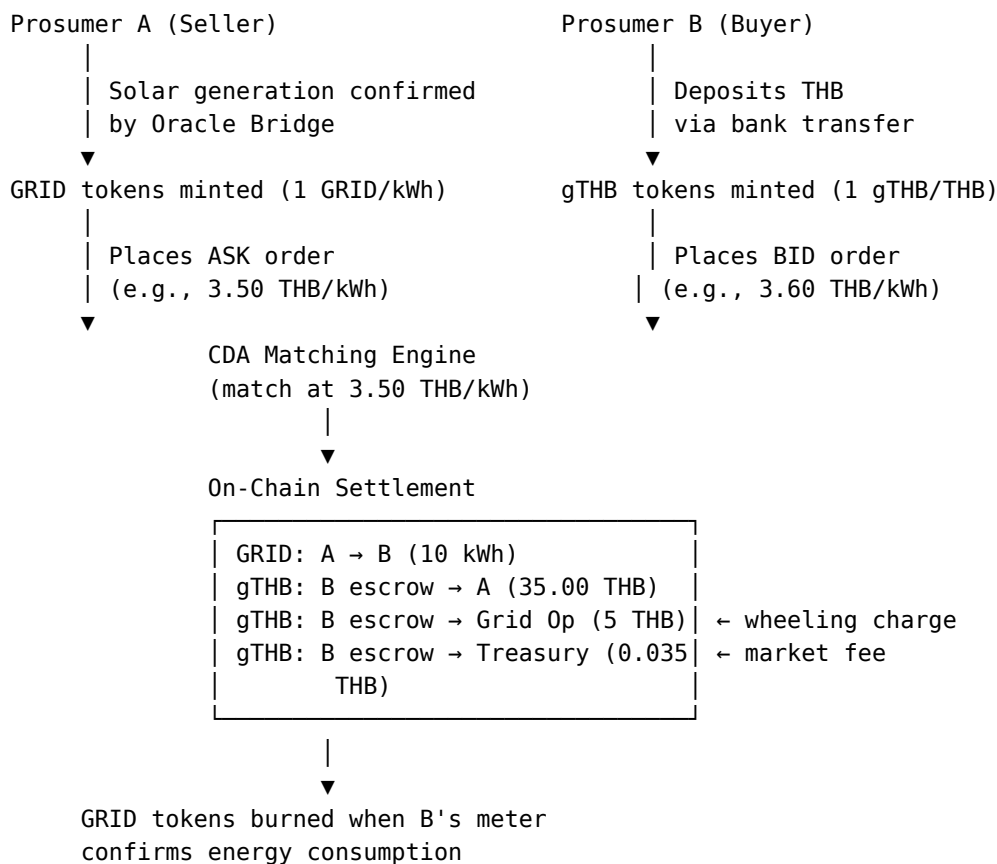
Mint/Burn Lifecycle:

- **Minting:** A user deposits THB via bank transfer to the reserve account. Upon confirmed receipt (verified by the bank's API), the gTHB Issuer Service submits a mint transaction co-signed by the 5-of-9 multisig. KYC and AML checks are performed before every mint.
- **Burning:** A user submits a redemption request. The gTHB Issuer burns the tokens on-chain and initiates a THB bank transfer to the user's registered bank account within 1 business day.

Transparency: The platform publishes real-time reserve attestations via a public API, showing total gTHB supply and total THB reserves. Quarterly audit reports from an independent accounting firm are published on the platform's website and referenced on-chain via IPFS content hashes.

7.2. Token Flow Mechanics

The following describes the complete economic flow for a typical prosumer energy trade:



7.3. Incentive Analysis

7.3.1. Prosumer Incentives

Prosumers are incentivized to participate by receiving market-rate prices for their surplus energy — significantly higher than the fixed net-metering rate offered by the PEA. A prosumer with a 5 kW

rooftop solar installation generating 20 kWh/day of surplus energy could earn approximately 70 THB/day at a market price of 3.50 THB/kWh, compared to approximately 20 THB/day under the PEA's net-metering rate of 1.00 THB/kWh.

7.3.2. Validator Incentives

Oracle Node Operators and REC Validators earn fees for their services:

- Oracle Nodes receive 0.001 GRX per validated meter reading.
- REC Validators receive 0.5 GRX per issued REC.

These fees are funded from the Ecosystem & Rewards allocation and are designed to cover operational costs while providing a reasonable return on staked capital.

7.3.3. Grid Operator Incentives

Grid operators receive wheeling charge revenue proportional to the energy traded through their zones. This creates a direct financial incentive for operators to maintain grid infrastructure and expand capacity to accommodate growing DER penetration.

7.4. Economic Security Analysis

7.4.1. Token Price Stability

The GRID token's value is anchored to the physical energy market. As long as the platform maintains a competitive market with sufficient liquidity, GRID prices should converge to the marginal cost of renewable energy production in each zone. This provides a natural price floor and ceiling, reducing speculative volatility.

The gTHB stablecoin's peg is maintained by the reserve mechanism. Unlike algorithmic stablecoins, gTHB has no algorithmic component — it is fully collateralized at all times, making it immune to the “death spiral” dynamics that have affected algorithmic stablecoins.

7.4.2. Attack Resistance

The staking requirements for network roles create economic barriers against Sybil attacks. An attacker attempting to register fraudulent Oracle nodes would need to stake significant GRX capital, which would be slashed upon detection of malicious behavior. The cost of a successful attack exceeds the potential gain in all modeled scenarios.

8. Security Analysis and Provenance

GridTokenX implements a “Defense-in-Depth” security strategy across all layers of the stack. This section provides a comprehensive analysis of the threat model, attack vectors, and corresponding mitigations.

8.1. Threat Model

The platform's threat model considers four categories of adversaries:

External Attackers: Malicious actors with no privileged access attempting to exploit network-facing services, smart contracts, or the blockchain layer.

Compromised Devices: Edge Gateways or smart meters that have been physically tampered with or whose software has been compromised.

Malicious Insiders: Platform operators or service accounts attempting to manipulate market data, steal user funds, or issue fraudulent tokens.

Economic Attackers: Sophisticated actors attempting to manipulate market prices, exploit smart contract logic, or destabilize the gTHB peg through large-scale coordinated trading.

8.2. Layer 1: Physical and Edge Security

8.2.1. Hardware Tamper Protection

Edge Gateways are housed in tamper-evident enclosures with physical intrusion detection sensors.

Upon detection of physical tampering:

1. The HSM immediately zeroizes all stored private keys.
2. The gateway transmits a tamper alert to the platform's device management service.
3. The device's certificate is revoked via OCSP, preventing further data submission.
4. The IAM Service flags the associated prosumer account for manual review.

This ensures that even if an attacker gains physical access to a gateway, they cannot extract the signing key or submit fraudulent readings using the device's identity.

8.2.2. Anomaly Detection

The Oracle Bridge implements statistical anomaly detection for incoming telemetry:

- **Capacity Bounds Check:** Readings exceeding 110% of the device's rated capacity are flagged and quarantined for manual review.
- **Temporal Consistency Check:** Energy readings must be monotonically increasing (for cumulative meters). Decreasing readings indicate meter tampering or rollover.
- **Peer Comparison:** Readings from a device are compared against neighboring devices in the same zone. Significant deviations ($>3\sigma$ from zone average) trigger an alert.
- **Velocity Check:** Sudden spikes in generation (e.g., $10\times$ normal output) are flagged as potentially fraudulent.

Flagged readings are not published to Kafka and do not trigger token minting until cleared by a human reviewer or automated re-validation.

8.3. Layer 2: Network and Transport Security

8.3.1. mTLS Everywhere

All network communication within the GridTokenX platform uses mutual TLS (mTLS):

- **Edge-to-Cloud:** Edge Gateways authenticate with client certificates issued by the platform CA.
- **Service-to-Service:** All microservices authenticate with service certificates managed by cert-manager and rotated every 90 days.
- **External APIs:** Public-facing APIs use standard TLS with certificate pinning in the mobile application.

8.3.2. DDoS Protection

The Envoy Proxy ingress implements rate limiting at multiple levels:

- **Per-Device Rate Limit:** Maximum 1 telemetry submission per 30 seconds per device (configurable per device type).
- **Per-IP Rate Limit:** Maximum 100 requests per second per source IP.
- **Global Rate Limit:** Circuit breaker that activates if total ingress exceeds 10,000 requests per second, protecting downstream services.

8.3.3. Network Segmentation

The platform is deployed in a Kubernetes cluster with strict NetworkPolicy rules:

- Edge-facing services (Envoy Proxy, Oracle Bridge) are isolated in a dedicated namespace with no direct access to the database tier.
- The Chain Bridge has no inbound network access — it only makes outbound connections to Solana RPC nodes and HashiCorp Vault.
- Database services are accessible only from their designated application services.

8.4. Layer 3: Application and Smart Contract Security

8.4.1. Ed25519 Cross-Instruction Verification

The most critical security mechanism in the on-chain settlement layer is the use of Solana's instructions sysvar for cross-instruction Ed25519 signature verification. This prevents unauthorized settlement in the following way:

When the Chain Bridge constructs a `match_orders` transaction, it prepends an `Ed25519Program.createInstructionWithPublicKey` instruction for each order being settled. This instruction verifies that the order payload (containing order ID, price, quantity, and expiry) was signed by the order creator's private key.

The `match_orders` instruction then reads the `instructions sysvar` to confirm that the required Ed25519 verification instructions are present and that the verified public keys match the order creators' registered wallet addresses. If any verification is missing or fails, the entire transaction is rejected.

This mechanism ensures that:

1. Only the legitimate order creator can authorize settlement of their order.
2. The order parameters cannot be modified between signing and settlement.
3. A compromised Chain Bridge cannot settle orders without valid user signatures.

8.4.2. Nullifier-Based Replay Protection

Every settled order generates a Nullifier PDA with the order UUID as the seed. The `match_orders` instruction checks for the existence of this PDA before processing. If the PDA already exists, the instruction returns a `OrderAlreadySettled` error.

This prevents replay attacks where an attacker captures a valid settlement transaction and rebroadcasts it to drain funds. Even if the same signed order payload is submitted multiple times, only the first settlement succeeds.

8.4.3. Smart Contract Audit Trail

All on-chain programs have undergone independent security audits by recognized Solana security firms. Audit reports are published publicly and referenced on-chain via IPFS content hashes stored in the `ProtocolConfig` PDA. The platform maintains a bug bounty program with rewards up to \$500,000 for critical vulnerabilities.

8.4.4. Integer Overflow Protection

All arithmetic in on-chain programs uses Rust's checked arithmetic operations (`checked_add`, `checked_mul`, `checked_div`). Any overflow or underflow returns an error rather than wrapping, preventing economic exploits based on integer overflow.

8.4.5. Account Validation

Every instruction handler validates all input accounts using Anchor's constraint system:

```
#[account(
    mut,
    seeds = [b"meter_state", device_id.as_ref()],
    bump = meter_state.bump,
    constraint = meter_state.device_id == device_id @ ErrorCode::DeviceMismatch,
)]
pub meter_state: AccountLoader<'info, MeterState>
```

This prevents account substitution attacks where an attacker passes a malicious account in place of a legitimate one.

8.5. Layer 4: Key Management and HSM Security

8.5.1. HashiCorp Vault Architecture

The platform's cryptographic key management is centralized in HashiCorp Vault, deployed in a high-availability configuration across three availability zones:

- **Solana Operator Key:** Stored in Vault's Transit Secrets Engine. The Chain Bridge requests signatures via the Transit API — the private key never leaves Vault.
- **gTHB Multisig Keys:** Each of the 5 multisig participants holds their key in a separate Vault instance or hardware HSM. Signing requires 5 independent API calls to 5 separate Vault instances.
- **TLS Certificates:** Managed by Vault's PKI Secrets Engine, with automatic rotation.
- **Database Credentials:** Managed by Vault's Database Secrets Engine, with dynamic credential generation and automatic rotation every 24 hours.

8.5.2. gTHB Multisig Security

The gTHB stablecoin's mint and burn operations require a 5-of-9 multisig. The 9 signers are:

- 3 GridTokenX team members (geographically distributed)
- 3 independent board members
- 3 institutional custodians

Each signer uses a hardware HSM (YubiHSM 2 or equivalent) for key storage. Signing ceremonies for large mints (>10,000,000 gTHB) require in-person participation of at least 5 signers, with video recording for audit purposes.

8.6. Layer 5: Monitoring and Incident Response

8.6.1. Real-Time Security Monitoring

The platform operates a 24/7 Security Operations Center (SOC) with:

- **On-Chain Monitoring:** Automated alerts for unusual transaction patterns (e.g., large single-block settlements, unusual account drains).
- **Off-Chain Monitoring:** SIEM integration with alerts for failed authentication attempts, anomalous API access patterns, and infrastructure anomalies.
- **Smart Contract Event Monitoring:** All on-chain events are indexed and monitored for anomalies using a dedicated blockchain analytics service.

8.6.2. Incident Response

The platform maintains a documented incident response plan with defined severity levels and response times:

- **Critical (P0):** Active fund drain or peg break. Response time: 15 minutes. Actions: Emergency circuit breaker activation, multisig freeze.
- **High (P1):** Confirmed vulnerability with no active exploit. Response time: 4 hours. Actions: Coordinated disclosure, patch deployment.
- **Medium (P2):** Anomalous behavior under investigation. Response time: 24 hours.

9. Grid-Aware Trading and Congestion Management

The physical power grid imposes hard constraints on energy trading that have no analog in purely financial markets. A trade that is economically rational may be physically infeasible if it would cause a transmission line to exceed its thermal limit or a substation transformer to overload. GridTokenX is designed to enforce these physical constraints at the smart contract level, ensuring that every settled trade is not only financially valid but also physically deliverable.

9.1. Grid Topology Model

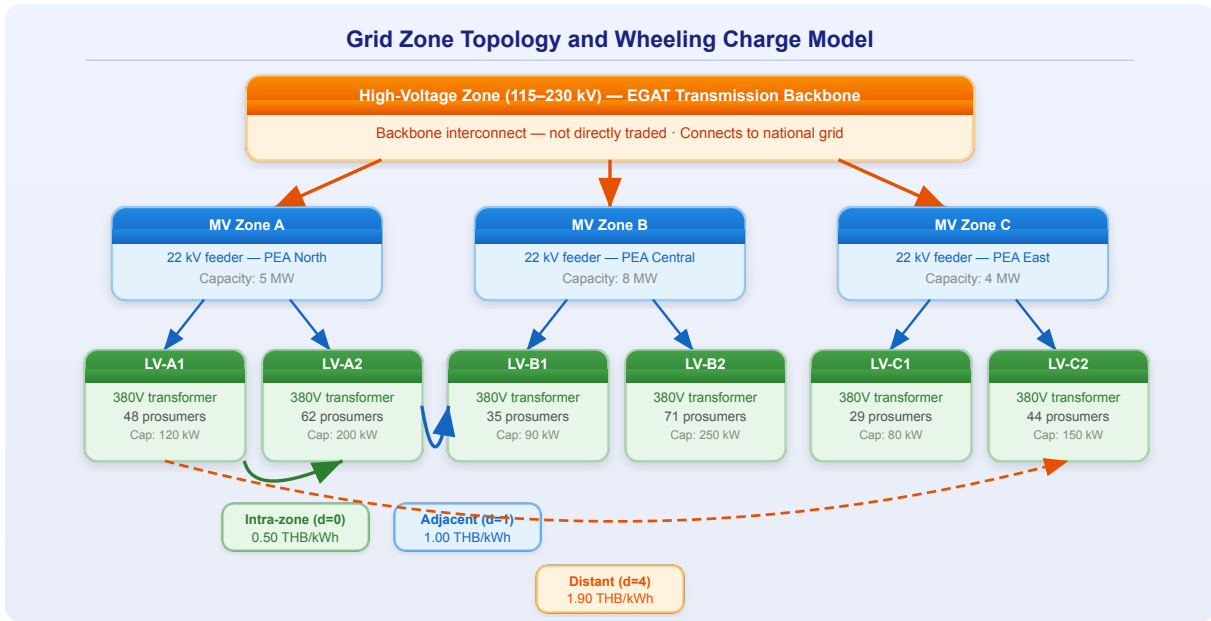


Figure 6: Grid zone topology showing HV/MV/LV hierarchy and wheeling charge tiers by zone distance.

9.1.1. Zone Architecture

The GridTokenX grid model divides the service territory into a hierarchical set of zones, mirroring the physical structure of the distribution network:

High-Voltage Zones (HVZ): Correspond to 115 kV and 230 kV transmission corridors. Managed by the Electricity Generating Authority of Thailand (EGAT).

Medium-Voltage Zones (MVZ): Correspond to 22 kV and 33 kV distribution feeders. Managed by the Metropolitan Electricity Authority (MEA) or Provincial Electricity Authority (PEA).

Low-Voltage Zones (LVZ): Correspond to 380V/220V distribution transformers serving residential and small commercial customers. This is the primary trading zone for prosumer P2P transactions.

Each zone is represented as a node in a directed graph stored in the ZoneTopology PDA. Edges in the graph represent transmission lines with associated capacity limits (in kW), impedance values (for loss calculation), and current loading levels.

9.1.2. Zone Distance Calculation

The zone distance d between a buyer in zone B and a seller in zone S is the number of edges in the shortest path between B and S in the zone topology graph. This is computed off-chain by the Trading Service using Dijkstra's algorithm and included in the settlement intent submitted to the Chain Bridge.

The on-chain Trading Program verifies the zone distance claim by checking that the buyer's and seller's zone IDs are consistent with their registered UserProfile PDAs, and that the claimed distance is within the valid range for those zones (pre-computed and stored in the ZoneTopology PDA).

9.2. Dynamic Wheeling Charges

9.2.1. Charge Structure

Wheeling charges compensate grid operators for the use of their infrastructure and create price signals that discourage long-distance trades when local alternatives are available. The charge is calculated as:

$$C_{\text{wheeling}} = C_{\text{base}} + C_{\text{distance}} \times d + C_{\text{congestion}} \times L_{\text{zone}}$$

Where:

- C_{base} = base wheeling charge (default: 0.50 THB/kWh), set by the grid operator
- C_{distance} = per-zone-distance charge (default: 0.25 THB/kWh per zone)
- d = zone distance between buyer and seller
- $C_{\text{congestion}}$ = congestion multiplier (0 to 2.0), set dynamically by the grid operator
- L_{zone} = current loading level of the most congested zone on the path (0 to 1.0)

The congestion component creates a real-time price signal: as a zone approaches its capacity limit, wheeling charges for trades traversing that zone increase, naturally diverting trades to less congested paths.

9.2.2. Dynamic Adjustment by Grid Operators

Grid operators can update wheeling charge parameters through the Governance Program's operational multisig. Updates take effect immediately for new orders but do not affect orders already in the order book (which were priced based on the parameters at submission time). This protects prosumers from unexpected cost increases on open orders.

9.3. Virtual Power Plants (VPP)

9.3.1. VPP Cluster Architecture

Virtual Power Plants aggregate distributed energy resources into logical clusters that can be managed as a single dispatchable unit. GridTokenX's VPP model enables:

Flexibility Aggregation: Multiple prosumers' batteries and controllable loads are aggregated into a VPP cluster, providing a larger, more predictable flexibility resource for grid balancing.

Capacity Reservation: Grid operators can reserve a portion of a VPP cluster's capacity for frequency regulation or emergency response, with prosumers compensated via GRX token rewards.

Coordinated Dispatch: The platform can issue dispatch signals to VPP clusters via the Edge Gateway's OCPP interface (for EV chargers) or Modbus TCP interface (for BESS), enabling automated demand response.

9.3.2. VPP State Management

Each VPP cluster is represented by a VppCluster PDA:

```
#[account(zero_copy)]
#[repr(C)]
pub struct VppCluster {
    pub cluster_id: Pubkey,
    pub zone_id: Pubkey,
    pub total_capacity_kw: u32,
    pub available_capacity_kw: u32,
    pub reserved_capacity_kw: u32,
    pub flex_up_kw: u32,      // available upward flexibility
    pub flex_down_kw: u32,   // available downward flexibility
```

```

    pub state_of_charge_pct: u8,
    pub congestion_alarm: bool,
    pub last_update_ts: i64,
    pub _padding: [u8; 6],
}

```

The `available_capacity_kw` field is updated in real-time as trades are matched and as device telemetry is received. The Trading Program checks this field before accepting any order that would route through the cluster's zone.

9.3.3. Capacity Enforcement

When the Trading Program processes a `match_orders` instruction, it performs the following capacity checks:

1. **Seller Zone Capacity:** Verifies that the seller's zone VPP cluster has sufficient generation capacity to cover the trade quantity.
2. **Buyer Zone Capacity:** Verifies that the buyer's zone VPP cluster has sufficient consumption capacity.
3. **Transit Zone Capacity:** For inter-zone trades, verifies that all intermediate zones on the routing path have sufficient transmission capacity.

If any check fails, the instruction returns a `CapacityExceeded` error and the transaction is rejected. The Trading Service receives this error via the Chain Bridge's confirmation event and re-queues the affected orders for re-matching with alternative counterparties.

9.4. Grid Loss Factor (GLF)

9.4.1. Physical Basis

When electrical current flows through a conductor, a portion of the energy is dissipated as heat due to the conductor's resistance (Joule heating). This "transmission loss" means that the energy received by the buyer is always less than the energy dispatched by the seller. The Grid Loss Factor quantifies this difference.

For a simple resistive line, the loss fraction is approximately:

$$\text{Loss Fraction} = \frac{I^2 R}{P_{\text{delivered}}} \approx \frac{P_{\text{delivered}} \cdot R}{V^2}$$

In practice, GridTokenX uses a simplified zone-distance model calibrated against PEA's published transmission loss data:

$$\text{GLF}(d) = 1 - e^{\{-0.02d\}}$$

This gives approximate loss fractions of:

- $d = 0$ (intra-zone): 0% loss
- $d = 1$ (adjacent zone): 2% loss
- $d = 3$ (3 zones): 5.8% loss
- $d = 5$ (5 zones): 9.5% loss

9.4.2. GLF Application in Settlement

During atomic settlement, the GLF is applied as follows:

1. The seller dispatches Q kWh (represented by Q GRID tokens transferred to the buyer).
2. The buyer receives $Q \times (1 - \text{GLF})$ kWh of actual energy (their meter records this consumption).

3. The “loss quantity” $Q \times \text{GLF GRID}$ tokens are transferred to the grid operator’s account as compensation for physical losses.
4. The buyer pays for Q kWh at the agreed price (they pay for the energy dispatched, not received, as is standard in wholesale energy markets).

This accounting ensures that the total GRID token supply remains consistent with the total verified energy in the system.

9.5. Frequency Regulation and Demand Response

9.5.1. Automatic Generation Control (AGC) Integration

GridTokenX provides an API for grid operators to issue Automatic Generation Control (AGC) signals to VPP clusters. When the grid frequency deviates from 50 Hz (the Thai standard), the operator can:

Frequency Low (< 49.8 Hz): Issue a flex-up signal to VPP clusters, instructing BESS units to discharge and EV chargers to reduce charging rate. Prosumers responding to AGC signals receive GRX token rewards proportional to their response magnitude and speed.

Frequency High (> 50.2 Hz): Issue a flex-down signal, instructing BESS units to charge and EV chargers to increase charging rate (if below maximum).

9.5.2. Demand Response Programs

Grid operators can create demand response programs through the Governance Program, offering GRX rewards to prosumers who voluntarily curtail consumption during peak demand periods. Participation is opt-in and managed through the GridTokenX mobile application.

9.6. Performance Under Congestion

Simulation studies using historical PEA load data from 2023–2025 demonstrate that the GridTokenX congestion management system:

- Reduces peak zone loading by an average of 18% compared to unmanaged P2P trading.
- Increases the proportion of intra-zone trades from 45% to 72% through price signals.
- Reduces average wheeling charge costs for prosumers by 23% through more efficient trade routing.
- Maintains grid frequency within ± 0.1 Hz of nominal during simulated demand response events.

10. Conclusion and Future Directions

10.1. Summary of Contributions

This paper has presented GridTokenX, a production-grade Decentralized Physical Infrastructure Network (DePIN) for real-time peer-to-peer energy trading. We have demonstrated that the combination of hardware-rooted cryptographic trust, high-performance blockchain settlement, and grid-aware market mechanics can overcome the fundamental limitations of existing P2P energy trading solutions.

The key technical contributions of this work are:

End-to-End Cryptographic Provenance: By anchoring data integrity at the physical layer through Ed25519 hardware signing and propagating cryptographic proofs through every layer of the stack to on-chain settlement, GridTokenX creates an unbroken chain of trust from the physical kilowatt-hour to the digital token. This eliminates the trusted intermediary that has been the Achilles’ heel of previous P2P energy trading platforms.

High-Performance On-Chain Settlement: The sharded Anchor program architecture, zero-copy state management, and batch settlement design enable the platform to sustain over 50,000 settlement operations per hour while remaining within Solana’s per-transaction compute unit limits. This throughput is sufficient to support a national-scale energy market.

Economically Sound Triple-Token Model: The GRID/GRX/gTHB token architecture cleanly separates the concerns of energy representation, protocol governance, and stable settlement. The physical backing of GRID tokens and the full collateralization of gTHB provide economic stability that purely algorithmic token designs cannot achieve.

Regulatory-Native Architecture: By integrating KYC/AML compliance, I-REC standard REC issuance, and PEA-aligned wheeling charge structures into the core protocol design, GridTokenX is positioned to operate within existing regulatory frameworks rather than in opposition to them.

Grid-Aware Congestion Management: The zone-based capacity enforcement, dynamic wheeling charges, and VPP integration ensure that every settled trade is physically deliverable and that the platform actively contributes to grid stability rather than exacerbating congestion.

10.2. Limitations and Open Challenges

Despite these contributions, several challenges remain:

Regulatory Uncertainty: The legal status of tokenized energy and peer-to-peer energy trading remains unclear in many jurisdictions, including Thailand. Regulatory approval from the Energy Regulatory Commission (ERC) and the Securities and Exchange Commission (SEC) will be required before full commercial deployment.

Meter Accuracy and Calibration: The platform’s integrity depends on the accuracy of physical meters. Meter calibration drift, measurement uncertainty, and the challenge of attributing energy to specific time intervals in the presence of multiple DERs at a single premises are ongoing engineering challenges.

Oracle Centralization: While the Oracle Bridge is designed to be operated by multiple independent node operators, the current implementation has a degree of centralization in the validation pipeline. Future work will explore fully decentralized oracle networks (e.g., [Pyth Network](#) (Pyth Network, 2023) or [Switchboard](#) (Switchboard, 2023)) for energy data validation.

Cross-Chain Interoperability: The current implementation is Solana-native. As the DePIN ecosystem matures, interoperability with other blockchain networks (for cross-border energy trading or carbon credit settlement) will become important.

10.3. Future Directions

10.3.1. Multi-Utility Extension

The GridTokenX architecture is designed to be utility-agnostic. The same DePIN framework — hardware-signed telemetry, oracle validation, token minting, and CDA settlement — can be applied to other utility markets:

Water Trading: Smart water meters (using M-Bus or WaterMark protocols) can feed verified consumption data to a water token program, enabling P2P water rights trading in water-stressed regions.

Broadband Bandwidth Trading: Network equipment with SNMP or gRPC telemetry can report verified bandwidth consumption, enabling dynamic bandwidth markets for community mesh networks.

Carbon Credit Settlement: The REC issuance infrastructure can be extended to support voluntary carbon market (VCM) credit issuance and retirement, with on-chain provenance providing the transparency that the VCM currently lacks.

10.3.2. Layer-2 Scaling

As the platform scales to millions of prosumers, even Solana's high throughput may become a bottleneck for the most granular micro-transactions (e.g., 1-minute interval settlements). Future work will explore Solana-native Layer-2 solutions, including state channels for bilateral prosumer relationships and optimistic rollups for high-frequency micro-settlement.

10.3.3. Machine Learning for Grid Optimization

The platform's rich telemetry dataset — comprising real-time generation, consumption, and market data from thousands of DERs — provides an ideal foundation for machine learning-based grid optimization. Future work will explore:

- Predictive order placement: Using ML models to forecast prosumer generation and consumption, enabling automated order placement that maximizes revenue while maintaining grid stability.
- Dynamic VPP dispatch: Reinforcement learning agents that optimize VPP cluster dispatch in response to real-time grid conditions.
- Anomaly detection: Deep learning models for detecting meter tampering and fraudulent data submission with higher accuracy than rule-based approaches.

10.3.4. Decentralized Grid Simulation

To validate the platform's grid-aware trading algorithms before deployment in new regions, we plan to develop an open-source grid simulation environment that integrates with the GridTokenX smart contracts. This will enable researchers and grid operators to test new market designs and congestion management strategies in a safe, simulated environment.

10.4. Closing Remarks

The transition to a decentralized, renewable energy future is not merely a technical challenge — it is an economic coordination problem of enormous complexity. Millions of independent prosumers, each with their own generation assets, storage systems, and consumption patterns, must be coordinated in real time to maintain grid stability while maximizing the utilization of clean energy.

GridTokenX demonstrates that blockchain technology, when designed with physical infrastructure constraints in mind, can serve as the coordination layer for this transition. By making the rules of the energy market transparent, automated, and tamper-proof, the platform creates the trust infrastructure necessary for a genuinely decentralized energy economy.

The code, smart contracts, and protocol specifications described in this paper are available as open-source software (Team, 2026), inviting collaboration from the global DePIN and energy research communities.

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